

# Tahsin R. Chowdhury

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## Education

### The University of Texas at Arlington

Aug 2019 - May 2024

BS Information Systems, STEM-Designated, AACSB | *Maverick Scholarship, Honor Roll, Dean's List* | Coursework: Full-Stack Development, Database Systems & BI, Data Structures & Algorithms, Statistics for Business Analytics, Cloud Computing, Information Security

## Technical Skills

**Languages:** JavaScript ES6+, TypeScript, Python, Java, SQL, HTML5, CSS3

**Frontend:** React 19, Next.js 15, Redux, Zustand, Tailwind CSS, GSAP, responsive design, WCAG 2.0

**Backend:** Node.js, Express, Python FastAPI, Java Spring Boot, REST APIs, OAuth 2.0, JWT, RBAC

**Data & Analytics:** PostgreSQL (CTEs, window functions, stored procs, RLS), MySQL, Power BI (DAX, Power Query, star schema), Tableau, pandas, NumPy, scikit-learn, Streamlit, Excel, ETL

**AI & ML:** LangGraph, LangChain, OpenAI API, Claude API, RAG pipelines, TensorFlow/Keras, prompt engineering, GitHub Copilot, Claude Code, Cursor

**Cloud & Tools:** AWS (Lambda, API Gateway, S3, CloudFront, IAM), Docker, Kubernetes, GitHub Actions, Vercel CI/CD, Git, Jest, RTL, Zod, Figma, Agile/Scrum

## Experience

### Self-Employed -- Arlington, TX — Founder & Full Stack Developer

Jan 2024 - Present

- Shipped 6 production full-stack applications end to end (React 19, Next.js 15, TypeScript, Node.js, Java Spring Boot, Python FastAPI, PostgreSQL, AWS) -- OAuth 2.0, JWT, RLS, RBAC, Jest/RTL TDD, and Vercel CI/CD across every project.
- Built a LangGraph stateful multi-actor AI agent, a RAG document-analysis pipeline, OpenAI function-calling integrations, and a TensorFlow/Keras image classifier; resolved a production frame-rate bottleneck via Chrome DevTools by offloading clustering across 300+ concurrent data points to a Web Worker.

### University of Texas at Arlington -- Arlington, TX — Software Developer

Apr 2021 - May 2024

- Led 4 Figma-to-code full-stack builds in Agile/Scrum teams -- architected a serverless AWS backend (Lambda, API Gateway, DynamoDB, RDS, S3, CloudFront, IAM) and integrated 6+ RESTful APIs with async state management, error boundaries, and retry logic.

### University of Texas at Arlington -- Arlington, TX — Data Analyst

Aug 2019 - Apr 2021

- Queried a normalized SQL database of 1,000+ employee records (joins, aggregations, stored procedures) to track payroll accuracy and headcount -- replaced manual Excel workflows, cutting reporting time 30%; built Tableau turnover and payroll dashboards for weekly reviews and Python scripts to clean raw HR data.

### Chowdhury Group -- Remote — IT Support Intern

Jan 2020 - Dec 2021

- Consolidated 200+ staff records and equipment assignments from separate Excel files into a single SQL database; automated weekly ticket reports with Python and built a Tableau volume dashboard that helped cut repeat tickets by an estimated 25%.

## Projects

**FUN - Real-Time Sports Platform** | [Live Demo](#) | [GitHub](#) — React 19, TypeScript, Node.js, Java Spring Boot, PostgreSQL, Supabase Realtime, Mapbox GL, Docker, AWS

- Geo venue discovery, atomic PostgreSQL RPC booking, and real-time messaging -- Web Worker clustering eliminated frame drops across 300+ concurrent data points.

**Skintrix AI** | [Live Demo](#) | [GitHub](#) — React 19, Next.js, TypeScript, Python FastAPI, ML microservice REST APIs, GSAP

- Pixel-accurate Figma-to-code diagnostic UI -- multi-step async ML response handling, confidence-score parsing, 50+ GSAP sequences, WCAG 2.0 compliant.

**Nike E-Commerce** | [Live Demo](#) | [GitHub](#) — Next.js 15, PostgreSQL, Drizzle ORM, Stripe, OAuth 2.0, Zod, Zustand

- SSR/CSR rendering boundaries, Stripe REST API, JWT-secured OAuth 2.0, Zod payload validation, and Vercel CI/CD.

**Summarist - SaaS Platform** | [Live Demo](#) | [GitHub](#) — Next.js 15, Firebase Auth, Stripe, Firestore

- RBAC premium content gated from Firebase Auth token through SSR boundaries -- Stripe billing, real-time Firestore sync, custom HTML5 audio player.

**AI SaaS Suite - Generative AI Apps** | [GitHub](#) — OpenAI API, Claude API, Gemini, LangChain, LangGraph, RAG

- HITL interview-prep tool, AI notebook, and portfolio Q&A agent -- multi-agent workflows, vector-search RAG, streaming responses, prompt engineering.

**NFT Ultraverse** | [Live Demo](#) | [GitHub](#) — React 19, Next.js, REST APIs, Tailwind CSS

- Figma-built NFT marketplace frontend -- dynamic REST data, collection filtering and search, optimized image loading.

**Retail & Library SQL Analysis** | [Live Demo](#) | [GitHub](#) — PostgreSQL, CTEs, window functions, stored procedures, CTAS

- Retail analysis on 1,000+ transactions (RANK/LAG/PARTITION BY, MoM trends, profit margins); 6-table library system with checkout/return stored procedures and overdue segmentation.

**Data Professional Survey Dashboard** | [Live Demo](#) | [GitHub](#) — Power BI, DAX, Power Query, star schema, ETL

- 630-respondent global survey -- Power Query ETL (salary parsing, null handling), star schema with DAX measures, treemap, gauge, and donut visuals.

**Data Science Web Apps** | [Live Demo](#) | [GitHub](#) — Python, Streamlit, pandas, NumPy, scikit-learn, Seaborn, Altair

- NBA stats explorer (web scraping, Seaborn heatmap), DNA nucleotide counter, stock and S&P 500 EDA apps, Iris/Penguins classifiers; 7K-customer churn cohort analysis in progress.